

# Olist Ecommerce Customer Satisfaction & Retention Analysis

This project focused on identifying the major drivers of customer dissatisfaction, operational inefficiencies, and customer retention behavior in the Olist ecommerce dataset. The analysis was conducted using Python for data cleaning and feature engineering, followed by SQL-based analytical investigation for business insights.

## 1. Business Questions

- What are the major reasons behind customer dissatisfaction?
- Do delayed deliveries reduce customer retention and repeat purchases?

## 2. Data Preparation & Engineering

- Data cleaning and preprocessing were performed in Python.
- Feature engineering included creating columns such as: Delay, Late, Late\_Delay, Review Categories, Delay Buckets, Seller Volume Segments, and Bad Review Indicators.
- SQL Server was used for business analysis and relationship investigation across customers, sellers, orders, and products.

## 3. Customer Dissatisfaction Analysis

- Delivery delay was identified as the strongest driver of customer dissatisfaction.
- Orders delivered within 1–2 days of delay had relatively moderate dissatisfaction levels.
- Customer dissatisfaction increased sharply after delays exceeded 3 days.
- Orders delayed by 5+ days generated extremely high bad review percentages.

## 4. Delay Severity Threshold Analysis

| Delay Bucket    | Bad Review Trend               | Average Review Score |
|-----------------|--------------------------------|----------------------|
| On Time / Early | Low dissatisfaction            | 4.29                 |
| 1–2 Days Delay  | Moderate dissatisfaction       | 3.91                 |
| 3–5 Days Delay  | Sharp dissatisfaction increase | 2.80                 |
| 5+ Days Delay   | Extremely high dissatisfaction | 1.78                 |

A critical operational threshold was discovered: customer dissatisfaction escalates dramatically once delivery delays exceed 3 days. Bad reviews increased nearly 3x between the 1–2 day delay bucket and the 3–5 day delay bucket.

## 5. Product Category Analysis

- Furniture and home-related categories consistently received high bad review rates.
- Some product categories received bad reviews even without delivery delays.
- This indicates that customer dissatisfaction is not caused only by logistics.

- Potential reasons include product quality issues, expectation mismatch, damages, or packaging problems.

| Category Pattern          | Interpretation                                     |
|---------------------------|----------------------------------------------------|
| Bad reviews without delay | Likely product quality or expectation issue        |
| Bad reviews with delay    | Strong logistics and delivery issue                |
| Furniture categories      | Sensitive to both logistics and product experience |

## 6. Seller Performance Segmentation

Sellers were segmented using quantile-based segmentation (pd.qcut) into Low, Medium, and High-volume seller groups based on total order count.

| Seller Segment | Delay Rate | Bad Review Rate | Insight                                   |
|----------------|------------|-----------------|-------------------------------------------|
| Low            | 7.16%      | 13.53%          | Weakest operational performance           |
| Medium         | 6.47%      | 12.32%          | Best operational balance                  |
| High           | 6.59%      | 13.36%          | Large sellers maintain stable performance |

## 7. Customer Retention Analysis

A customer-level retention model was created to test whether delayed first-order experiences reduce future purchases. The analysis isolated the first customer order and compared repeat purchase behavior between delayed and non-delayed first-order experiences.

| First Order Experience | Total Customers | Repeat Customers | Repeat Rate |
|------------------------|-----------------|------------------|-------------|
| No Delay               | 86,460          | 2,610            | 3.02%       |
| Delayed First Order    | 6,218           | 164              | 2.64%       |

Customers whose first order was delayed were approximately 12–13% less likely to reorder. This confirms that poor first delivery experiences negatively impact long-term customer retention.

## 8. Final Business Conclusions

- Delivery delay is the primary driver of customer dissatisfaction.
- Customer dissatisfaction increases dramatically when delays exceed 3 days.
- Furniture and home-product categories experience high dissatisfaction both with and without delays.
- Seller operational quality has a moderate influence on customer experience.
- Delayed first-order experiences reduce repeat purchase probability.
- Both logistics performance and product quality improvements are required to improve customer satisfaction.

E-Commerce Customer Satisfaction & Retention Analysis



93K  
Total Customers



97K  
Orders



6.59%  
Delay



7.00  
Avg Delay



2.98%  
Repeat\_Rate



13.28%  
Bad Review



4.14  
Average Review

